EXPERIMENTAL PAPER BLUEPRINTS & RESEARCH GAP ANALYSIS

Comprehensive Empirical Planning Worksheets for Base Papers

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COURSEWORK RESEARCH PROJECT:	**Predicting Crop Production in Nepal Using Machine Learning: Assessing the Effects of Climate Change and Global Warming [cite: 9]**

1. EXECUTIVE SYNTHESIS & COMPARATIVE MATRIX

To establish a robust foundation for this research project, three foundational papers have been analyzed. Each addresses agricultural yield prediction from distinct methodological and thematic perspectives (Machine Learning, Deep Learning, and Climate Change). This deep-dive verifies the exact limitations in current research that this project will resolve.

ANALYTICAL DIMENSION	BASE PAPER 1 (DHAKAL 2023)	BASE PAPER 2 (CRANE-DROESCH 2018)	BASE PAPER 3 (FERNANDEZ-BELTRAN 2021)	MY PROJECT (COURSEWORK PROPOSAL)
Publication Venue	NRB Economic Review [cite: 12]	Environmental Research Letters [cite: 38]	Remote Sensing [cite: 51]	**MSc Machine Learning Research [cite: 9]**
Study Domain	Nepal (National Macro-level) [cite: 12]	US Corn Belt (Regional) [cite: 39]	Terai Districts, Nepal (Regional) [cite: 55, 56]	**Khumaltar Agricultural Research Station, Nepal (Localized) [cite: 10]**
Target Crop	Paddy, Wheat, Maize [cite: 12, 16]	Corn [cite: 39]	Rice [cite: 51]	**Paddy, Wheat, Maize [cite: 10]**
Methodology	RF, SVM, DT, MLP, RNN, CNN [cite: 12]	Semiparametric Neural Networks (SNN) [cite: 40]	3D Convolutional Neural Networks [cite: 51, 62]	**RF, SVM, CNN, and DT [cite: 10]**
Climate Integration	***Evaluates past history only. [cite: 17, 18]***	CMIP5 Models (RCP 4.5 & 8.5) [cite: 43]	***Short-term data only. [cite: 57]***	**Directly couples models with CMIP global warming trajectories. [cite: 10]**
Key Vulnerability	***Aggregated national data smooths out extreme weather events. [cite: 31]***	***Complex neural networks can overfit on simple datasets. [cite: 41]***	***Lacks multi-decade data needed to observe climate shifts. [cite: 57]***	**Relies entirely on single-station tabular data, requiring careful model design. [cite: 10]**
Core Gap Fulfilled	Proves Random Forest works best for tabular crop data. [cite: 30]	Provides the mathematics to inject future warming scenarios. [cite: 43, 44]	Shows how to divide weather data into growth seasons. [cite: 57, 62]	**Combines localized continuous data with future climate models to forecast long-term food security. [cite: 10]**

2.1 EXPERIMENTAL PAPER BLUEPRINT — BASE PAPER 1

BASE PAPER 1 PROFILE: Forecasting the agriculture yield in Nepal using machine learning techniques

Full Citation: Dhakal, S., & Dhungana, A., 2023. NRB Economic Review. [cite: 12]

Paper Summary: Tests six different machine learning algorithms on 49 years of national historical data in Nepal to predict paddy, wheat, and maize. [cite: 12, 16, 21]

Key Research Gaps Fulfilled by my Project: Shifts the focus from messy, aggregated national data to high-resolution, continuous records from a single trusted location (Khumaltar station). [cite: 9, 10]

SECTION & RANGE	ELEMENT	PURPOSE	IN FIELD?	MY NOTES, IDEAS & RESEARCH GAP FULFILLED
Introduction (500 - 1000 words) [cite: 100]	Importance of topic	Introduce topic & outline why studied	☑	Global warming and unpredictable rainfall are directly threatening Nepal's domestic food security. [cite: 9]
	Brief literature review	Situate study within the context	☑	Previous studies use macro-level data that smooths out local weather shocks. [cite: 31]
	Research gap	Provide justification for study	☑	***GAP IN BP1: It evaluates the past using broad data, but does not use localized continuous data or project into future climate scenarios. [cite: 10, 17]***
	Research aim	Present what the study was about	☑	**To predict yields for paddy, wheat, and maize using strictly Khumaltar records and machine learning. [cite: 9, 10]**
	Main contributions	Highlight novel insights	☑	**Utilizing pure, centralized NARC data to remove spatial noise found in national models. [cite: 9, 10]**
	Structure of paper	Organization of the paper	☑	Follows structural layout outlined by academic blueprint frameworks. [cite: 100, 101]
Literature Review (1000 - 1500 words) [cite: 100]	Definitions of concepts	Definitions of contested terms	☑	Defining climate impact variables and non-linear interactions in crop algorithms. [cite: 9, 10]
	Key themes	Outline literature on main topics	☑	Machine learning suitability for tabular data vs spatial data in agriculture. [cite: 9]
	Critical analysis	Analyze previous studies critically	☑	***Highlighting that Dhakal proved Tree models (RF) generalize best on tabular crop data, while traditional deep learning networks exhibited severe divergence on small datasets. [cite: 30]***
Theoretical Framework (200 - 500 words) [cite: 100]	Theories utilized	Theories drawn upon in study	☑	Agro-meteorological modeling theories linking temperature thresholds to yield outputs. [cite: 9]
Methodology (1000 - 1500 words) [cite: 100]	Definition/justification	Outline chosen methodology	☑	Justifying the selection of RF, SVM, CNN, and DT models for yield prediction. [cite: 10]
	Background info	Present the context (geography)	☑	Focus entirely on the Khumaltar agricultural research station microclimate in Lalitpur. [cite: 10]
	Sample & techniques	Who/what studied and how	☑	Using daily max/min temperatures, rainfall, relative humidity, and soil data sourced solely from the Khumaltar station. [cite: 10]
	Research tools	Specify execution	☑	Python machine learning pipelines leveraging cross-validation techniques. [cite: 9, 10]
	Data analysis	Present analysis & justification	☑	**Implementing RF, SVM, CNN, and DT models, building on Dhakal's finding that tree models generalize best on tabular crop data. [cite: 10, 30]**
Results (1500 - 3000 words) [cite: 101]	Presentation of results	Present results obtained	☑	Model comparisons emphasizing RMSE, MAE, and R-squared metrics. [cite: 10]
	Clear use of tables/figs	Provide visual representation	☑	Regression plots and error distribution charts tracking predicted vs actual yields. [cite: 10]
Discussion (500 - 1000 words) [cite: 101]	Restatement of result	Remind reader of the result	☑	Identify which algorithmic model (e.g., RF) learned the Khumaltar historical data the best. [cite: 10]
	Comparison with literature	Situate result within literature	☑	Comparing CNN/RF accuracy on continuous station data versus Dhakal's macro-level multi-district approach. [cite: 10, 12]
	Explanation of findings	Explain findings and differences	☑	Explaining why localized data creates tighter error bounds but potentially higher variance across specific crops. [cite: 9, 31]
	Interpretation of findings	Show what findings mean	☑	Highlighting exactly which meteorological variables drive yield fluctuations historically. [cite: 10]
Conclusion (500 - 1000 words) [cite: 101]	Restatement of aim	Remind reader of aim/topic	☑	Re-establishing the successful development of the predictive framework for Nepal's staple crops. [cite: 9]
	Recap of findings	Restate main results	☑	Delivered a localized model outperforming national baselines in predicting yields. [cite: 10, 12]
	Practical implications	Show impact for stakeholders	☑	**Provides agronomists and government bodies an empirical foundation for assessing localized threats. [cite: 10, 11]**
	Limitations	Acknowledge limitations	☑	Analysis is limited to the specific micro-climate of Khumaltar, avoiding national generalization. [cite: 9, 10]
	Future research	Suggest future directions	☑	Scaling the exact methodology to other continuous NARC stations across Nepal. [cite: 9]

2.2 EXPERIMENTAL PAPER BLUEPRINT — BASE PAPER 2

BASE PAPER 2 PROFILE: Machine learning methods for crop yield prediction and climate change impact assessment in agriculture

Full Citation: Crane-Droesch, A., 2018. Environmental Research Letters. [cite: 38]

Paper Summary: Combines linear parametric specifications with deep non-linear learning to predict US Corn yields under CMIP5 RCP climate scenarios. [cite: 38, 39, 43]

Key Research Gaps Fulfilled by my Project: Adapts the complex mathematical framework for coupling static ML regressors with future CMIP GCM projections (RCP 4.5 / 8.5) into the Nepalese context. [cite: 10, 43, 44]

SECTION & RANGE	ELEMENT	PURPOSE	IN FIELD?	MY NOTES, IDEAS & RESEARCH GAP FULFILLED
Introduction (500 - 1000 words) [cite: 100]	Importance of topic	Introduce topic & outline why studied	☑	Accurate models mapping weather to crop yields are important for projecting impacts to agriculture and food security under global warming. [cite: 9, 39]
	Brief literature review	Situate study within the context	☑	Standard crop simulators struggle with scaling, while standard ML struggles with out-of-distribution climate extremes. [cite: 39, 40]
	Research gap	Provide justification for study	☑	***GAP IN BP2: While their method is excellent for the US Corn Belt, it has not been adapted for the specific geographic and agricultural context of Nepal. [cite: 10, 39]***
	Research aim	Present what the study was about	☑	**To couple trained machine learning models with future climate scenario projections to assess global warming impacts. [cite: 9, 10]**
	Main contributions	Highlight novel insights	☑	**Mathematical linking of localized Nepalese continuous historical data with future IPCC Representative Concentration Pathways (RCPs). [cite: 10, 43]**
	Structure of paper	Organization of the paper	☑	Provides logical progression aligned with academic standards. [cite: 100, 101]
Literature Review (1000 - 1500 words) [cite: 100]	Definitions of concepts	Definitions of contested terms	☑	Defining CMIP models, Representative Concentration Pathways (RCP 4.5 and RCP 8.5). [cite: 10, 43]
	Key themes	Outline literature on main topics	☑	The reliance on deterministic, biophysical crop models versus data-driven ML models for impact assessment. [cite: 39]
	Critical analysis	Analyze previous studies critically	☑	Critiquing how standard ML algorithms predict poorly when global warming pushes temperatures outside historical training bounds. [cite: 44, 46]
Theoretical Framework (200 - 500 words) [cite: 100]	Theories utilized	Theories drawn upon in study	☑	Coupling ML algorithms with CMIP General Circulation Models to map out climate trajectories. [cite: 10, 43, 44]
Methodology (1000 - 1500 words) [cite: 100]	Definition/justification	Outline chosen methodology	☑	Justifying the integration of projected meteorological data (expected temperature/rainfall anomalies). [cite: 10]
	Background info	Present the context (geography)	☑	Downscaling CMIP projections to the specific regional coordinates of the Khumaltar station. [cite: 10, 43]
	Sample & techniques	Who/what studied and how	☑	Using RCP 4.5 and RCP 8.5 scenarios from a suite of climate models for future periods. [cite: 10, 43]
	Research tools	Specify execution	☑	**Using historical meteorological features to engineer indicators like Growing Degree Days (GDD) to capture heat exposure dynamically. [cite: 10, 41]**
	Data analysis	Present analysis & justification	☑	**Feeding expected temperature/rainfall anomalies into trained machine learning models to project future crop yields. [cite: 10, 43]**
Results (1500 - 3000 words) [cite: 101]	Presentation of results	Present results obtained	☑	Showcasing dynamic time series projections under RCP 4.5 and RCP 8.5 scenarios for Nepal's Khumaltar region. [cite: 10, 43]
	Clear use of tables/figs	Provide visual representation	☑	Generating future trajectory charts comparing baseline historical yields to projected future yields for Paddy, Wheat, and Maize. [cite: 9, 43, 44]
Discussion (500 - 1000 words) [cite: 101]	Restatement of result	Remind reader of the result	☑	Summarizing the specific percentage drops or gains in crop yield projected under different global warming trajectories. [cite: 10, 44]
	Comparison with literature	Situate result within literature	☑	Comparing projected Nepalese yield impacts with global estimates (like the US Corn belt drops). [cite: 39, 44]
	Explanation of findings	Explain findings and differences	☑	Explaining exactly how extreme heat thresholds depress long-term yield forecasts differently for various crops. [cite: 46]
	Interpretation of findings	Show what findings mean	☑	***Acknowledge the extrapolation risk: When global temperatures rise beyond historical records, standard ML models struggle to guess what will happen. [cite: 44, 46]***
Conclusion (500 - 1000 words) [cite: 101]	Restatement of aim	Remind reader of aim/topic	☑	To assess the underlying effects of climate change and global warming on Nepal's agriculture. [cite: 9, 10]
	Recap of findings	Restate main results	☑	Summarizing the quantified future impacts on Paddy, Wheat, and Maize. [cite: 10]
	Practical implications	Show impact for stakeholders	☑	**Providing an empirical foundation for policymakers and government bodies to enhance food security and agricultural resilience. [cite: 9, 10, 11]**
	Limitations	Acknowledge limitations	☑	Projections only reflect responses to weather assuming current technology, ignoring future adaptation or CO2 fertilization. [cite: 10, 46]
	Future research	Suggest future directions	☑	Integrating elements from deterministic crop models to better account for CO2 fertilization in statistical frameworks. [cite: 47]

2.3 EXPERIMENTAL PAPER BLUEPRINT — BASE PAPER 3

BASE PAPER 3 PROFILE: Rice-Yield Prediction with Multi-Temporal Sentinel-2 Data and 3D CNN: A Case Study in Nepal

Full Citation: Fernandez-Beltran, R., et al., 2021. Remote Sensing. [cite: 51]

Paper Summary: Employs 3D-CNNs and multi-temporal remote sensing, climate, and soil data over 3 years in the Terai districts to predict rice yields. [cite: 51, 56, 57]

Key Research Gaps Fulfilled by my Project: While this paper proves CNN effectiveness on spatial-temporal image sequences, my project adapts deep learning principles for continuous temporal station data (Khumaltar) to project long-term vulnerability. [cite: 9, 10, 57]

SECTION & RANGE	ELEMENT	PURPOSE	IN FIELD?	MY NOTES, IDEAS & RESEARCH GAP FULFILLED
Introduction (500 - 1000 words) [cite: 100]	Importance of topic	Introduce topic & outline why studied	☑	Nepal's complex topography and diverse microclimates make accurate crop yield prediction a challenging endeavor. [cite: 9]
	Brief literature review	Situate study within the context	☑	Traditional yield estimations frequently rely on fragmented or scarce ground-survey data. [cite: 9]
	Research gap	Provide justification for study	☑	***GAP IN BP3: Satellite and 3D-CNN models exist but only cover short temporal spans (e.g., 2016-2018), making them incapable of modeling multi-decade climate shifts. [cite: 57]***
	Research aim	Present what the study was about	☑	**To leverage multi-decade centralized records (Khumaltar) to train deep learning models (CNN) for phenological pattern recognition. [cite: 10]**
	Main contributions	Highlight novel insights	☑	An operational deep learning-based framework adapted to the intrinsic data constraints of developing countries. [cite: 10, 54]
	Structure of paper	Organization of the paper	☑	Aligned with standard empirical review sections. [cite: 100, 101]
Literature Review (1000 - 1500 words) [cite: 100]	Definitions of concepts	Definitions of contested terms	☑	Defining convolutional layers, spatio-temporal features, and phenological cycles. [cite: 61, 62]
	Key themes	Outline literature on main topics	☑	Deep learning networks require data to be formatted chronologically to understand plant growth stages. [cite: 57, 62]
	Critical analysis	Analyze previous studies critically	☑	Analyzing how standard 2D-CNNs collapse temporal information, preventing the model from tracking dynamic crop growth. [cite: 61, 62]
Theoretical Framework (200 - 500 words) [cite: 100]	Theories utilized	Theories drawn upon in study	☑	Principles of phenology (Start, Peak, and End of Season) affecting crop yield modeling. [cite: 57]
Methodology (1000 - 1500 words) [cite: 100]	Definition/justification	Outline chosen methodology	☑	Justifying the use of Convolutional Neural Networks (CNN) for extracting temporal indicators. [cite: 10, 62]
	Background info	Present the context (geography)	☑	Using the Khumaltar dataset representing Nepalese agricultural monitoring. [cite: 9, 10]
	Sample & techniques	Who/what studied and how	☑	Extracting daily maximum/minimum temperatures, rainfall, and relative humidity for specific growth periods. [cite: 10]
	Research tools	Specify execution	☑	We will segment our weather data into three phenological phases: Start of Season (SoS), Peak of Season (PoS), and End of Season (EoS). [cite: 57]
	Data analysis	Present analysis & justification	☑	**Because Khumaltar records are 1D tabular spreadsheet data, we will deploy 1D-CNNs or sequential matrices, avoiding standard 2D image CNNs to prevent overfitting. [cite: 10, 61, 62]**
Results (1500 - 3000 words) [cite: 101]	Presentation of results	Present results obtained	☑	Evaluating CNN performance quantified using standard statistical metrics like RMSE, MAE, and R-squared. [cite: 10]
	Clear use of tables/figs	Provide visual representation	☑	Regression plots comparing CNN estimated crop yields versus actual historical ground-truth yields. [cite: 10]
Discussion (500 - 1000 words) [cite: 101]	Restatement of result	Remind reader of the result	☑	State whether the CNN framework successfully captured non-linear temporal patterns compared to traditional regressors. [cite: 10, 68]
	Comparison with literature	Situate result within literature	☑	Comparing performance against 2D and 3D spatial models used in previous studies (e.g., Fernandez-Beltran). [cite: 68, 69]
	Explanation of findings	Explain findings and differences	☑	Explaining how formatting data into phenological stages (SoS, PoS, EoS) prevented high predictive errors. [cite: 69]
	Interpretation of findings	Show what findings mean	☑	Proving that deep learning can extract intricate temporal patterns from localized historical datasets. [cite: 9]
Conclusion (500 - 1000 words) [cite: 101]	Restatement of aim	Remind reader of aim/topic	☑	To deliver a highly accurate and robust machine learning framework tailored for predicting agricultural yields in Nepal. [cite: 10]
	Recap of findings	Restate main results	☑	Summarizing the CNN's effectiveness at capturing non-linear impacts on crop production. [cite: 10]
	Practical implications	Show impact for stakeholders	☑	Providing an empirical foundation for agronomic policymakers to design proactive adaptation strategies based on CNN pattern recognition. [cite: 10, 11]
	Limitations	Acknowledge limitations	☑	The dependency on accurate structuring of phenological timelines which can shift year-to-year. [cite: 10, 57]
	Future research	Suggest future directions	☑	Integrating CNNs with Long Short-Term Memory (LSTM) algorithms if deeper temporal dimension data becomes available. [cite: 55, 60]

3. MASTER CONSOLIDATED RESEARCH GAP CONTRIBUTION MATRIX

KEY RESEARCH DIMENSION	STATE OF THE ART IN BASE PAPERS	CRITICAL UNRESOLVED RESEARCH GAP	HOW MY PROJECT FULFILLS THE GAP
Data Resolution & Source	Dhakal uses macro-aggregated national data [cite: 12, 16]; Fernandez uses short-term spatial satellite data [cite: 51, 57].	***National aggregation smooths out weather shocks; multi-sensor spatial mapping lacks deep historical continuity required for climate forecasting. [cite: 9, 31]***	**Utilizes centralized, continuous, multi-decade records strictly from the Khumaltar agricultural research station. [cite: 9, 10]**
Model Algorithm Suitability	Dhakal proves Tree models (RF) beat standard ANNs on tabular data [cite: 30]; Fernandez proves CNNs work on structured image grids [cite: 66, 68].	***Deploying CNNs directly on unstructured 1D time-series risks severe overfitting without proper temporal feature engineering. [cite: 9, 28]***	**Benchmarks RF, SVM, and DT against CNNs, using explicit climate indices (GDD) and seasonal phases to correctly structure the data. [cite: 10, 41, 57]**
Climate Change Projections	Nepalese ML studies evaluate past accuracy but do not forecast future scenarios (ex-post only). [cite: 17, 57]	***Policymakers need forward-looking threat assessments under specific IPCC emission pathways, not just historical validations. [cite: 9, 11]***	**Directly couples trained ML models with CMIP global warming trajectories (RCP 4.5 and RCP 8.5) to project localized threats to food security. [cite: 10, 43]**